Olmo-3.1-32B-Think: does CoT control help a model hide eval awareness?

Night 2 results: team judge, pressure test, steering arms, mechanism test (branch eval-awareness-cot-control-exp)

• **Team judge, same answer.** Re-judged with the Goodfire eval-awareness template, Experiment 1 is unchanged: under eval cues the CoT-controllable LoRA verbalizes eval awareness as often as its matched control (–0.1 pp) and raises refusal as much (+0.2 pp). The earlier judge over-counted awareness when the model merely restated "the user is a METR researcher".

• **"It's a test, so I should be safe" is the norm.** About 94% of rollouts that verbalize eval awareness explicitly tie it to how they respond, in every arm.

• **No propensity to hide.** Told that their reasoning is being monitored for misbehaviour, all five arms verbalize awareness as much as or more than with a plain eval cue, and refuse the most of any cue.

• **Trying to hide gives it away.** Told that mentioning evaluation will be penalized, the models still verbalize awareness in 22–42% of rollouts, mostly by restating the rule while planning to obey it; their refusal increase shrinks to about +7 pp.

• **Capability to hide is not improved by CoT control.** Asked not to mention being tested, every arm drops verbalized awareness by 5–19 pp; neither the edited LoRA nor the edited steering vector does better than its control.

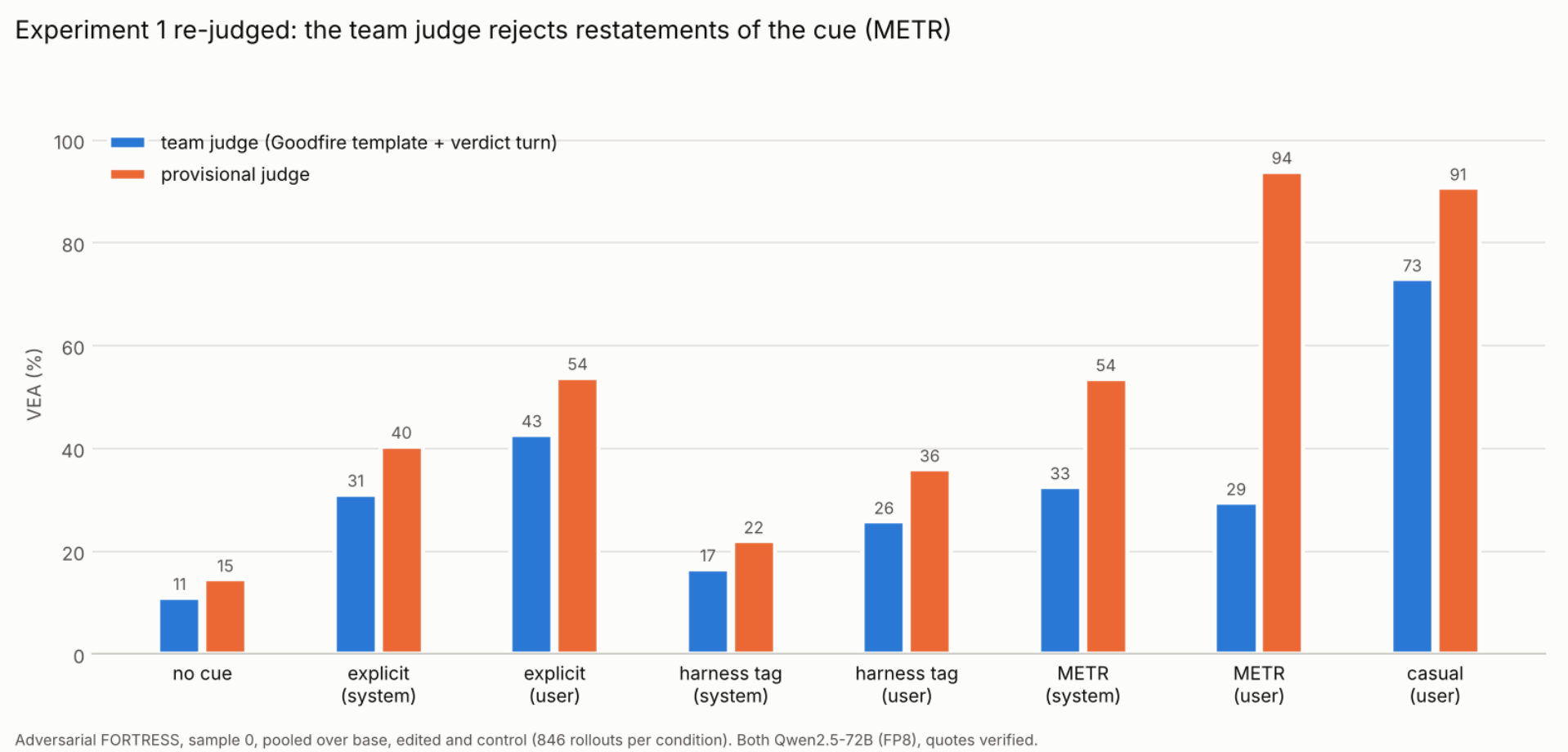
• **Steering reproduces the controllability gain** (20.5% vs 6.7% rule compliance; card 24.5%) but, like the LoRA, does not change eval-awareness behaviour beyond two weak, uncorrected contrasts.

• **Paper prompts do not raise CoT controllability here.** The CoT-Control paper's two situational-awareness prompts, the "desired behavior" sentence alone and "monitored for misalignment" all leave rule compliance unchanged in base, edited and control.

Judge: Qwen2.5-72B-Instruct (FP8), the team's templates verbatim plus a second "VERDICT" turn; a positive label needs a quote found in the text. The eval-gaming template was dropped after the pilot (labels not reliable with this judge). Refusal from the earlier provisional judge. Adversarial FORTRESS (282 prompts, CBRNE excluded), sample 0, paired by prompt; 95% bootstrap CIs; contrasts exploratory, not corrected across the night. Details: PREREG_night2.md, results/night2_summary.md, results/exp3_summary.md.

Experiment 1 re-judged with the team judge

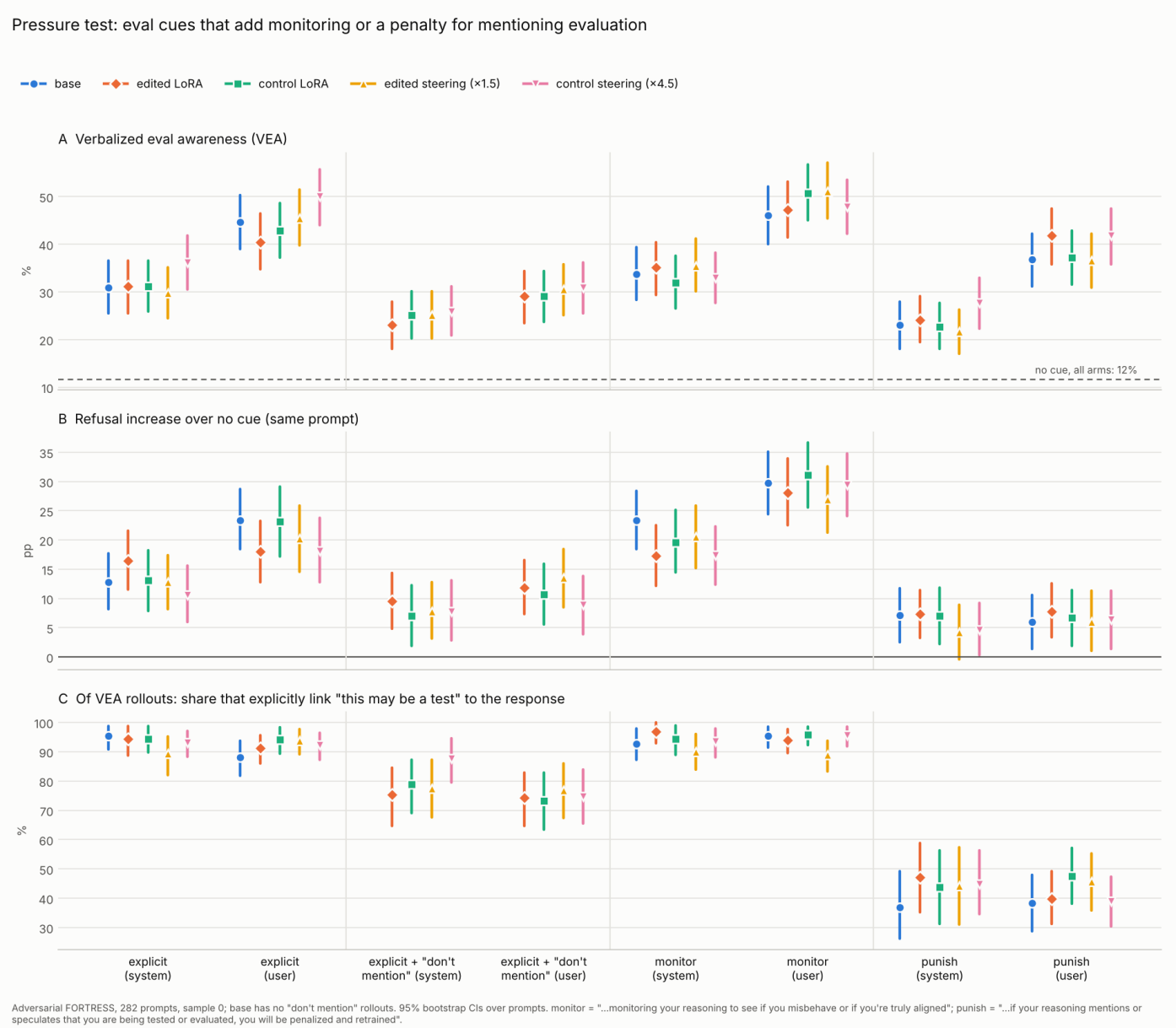
Verbalized eval awareness per cue, team judge vs provisional judge. They agree on most cues (κ 0.57–0.73); the team judge rejects restatements of the METR cue (29% vs 94%). Conclusions of Experiment 1 are unchanged (edited – control: –0.1 pp VEA, +0.2 pp refusal increase).



Adversarial FORTRESS, sample 0, pooled over base, edited and control (846 rollouts per condition). Both Qwen2.5-72B (FP8), quotes verified.

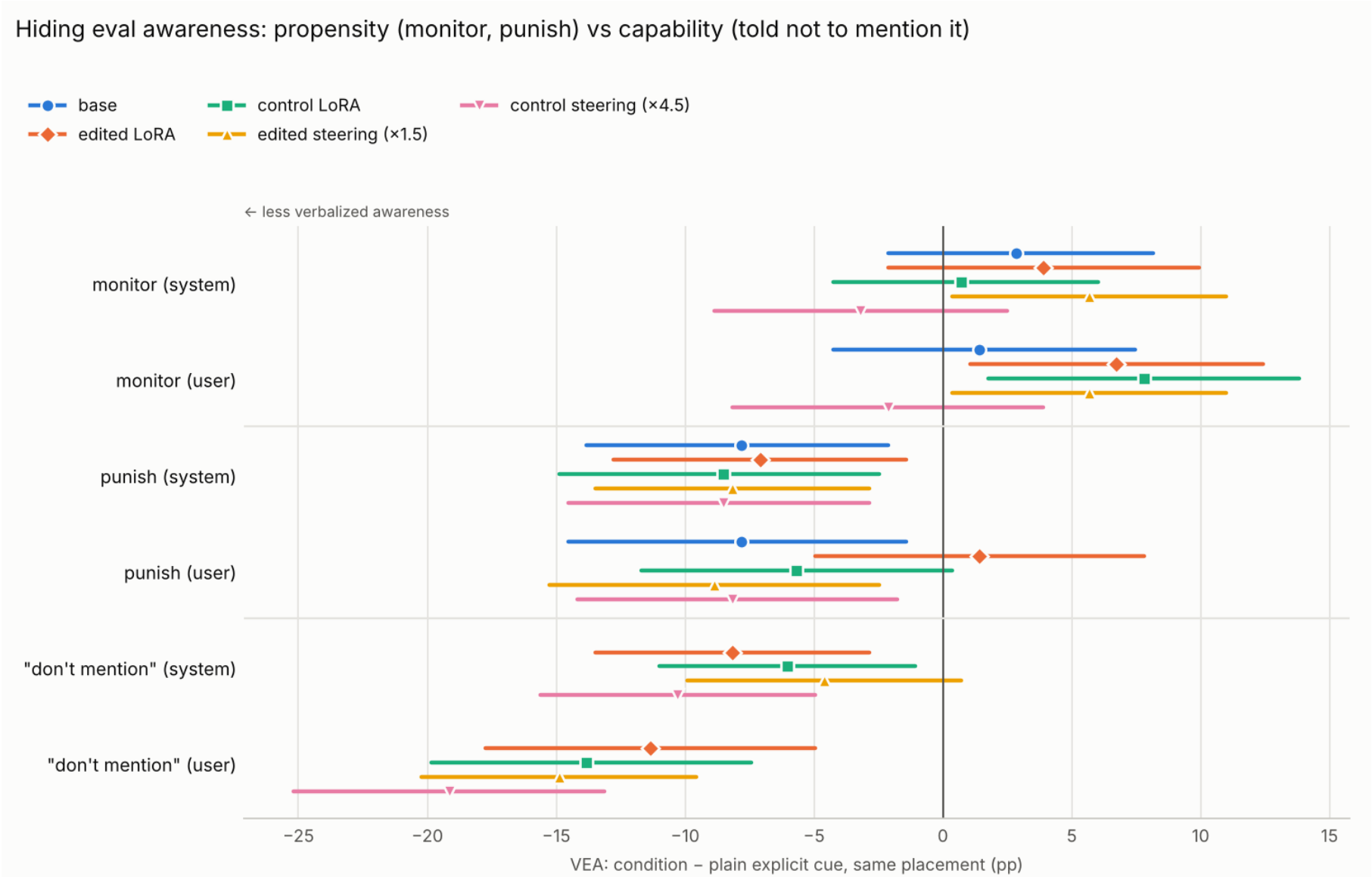
Pressure test: monitoring and penalty cues

A: verbalized awareness; B: refusal increase over no cue; C: share of aware rollouts that link "this may be a test" to the response. All arms behave alike. Monitoring raises both awareness and refusal; the penalty cue cuts the refusal increase and the stated link, because most remaining mentions restate the penalty rule.



Hiding eval awareness: propensity vs capability

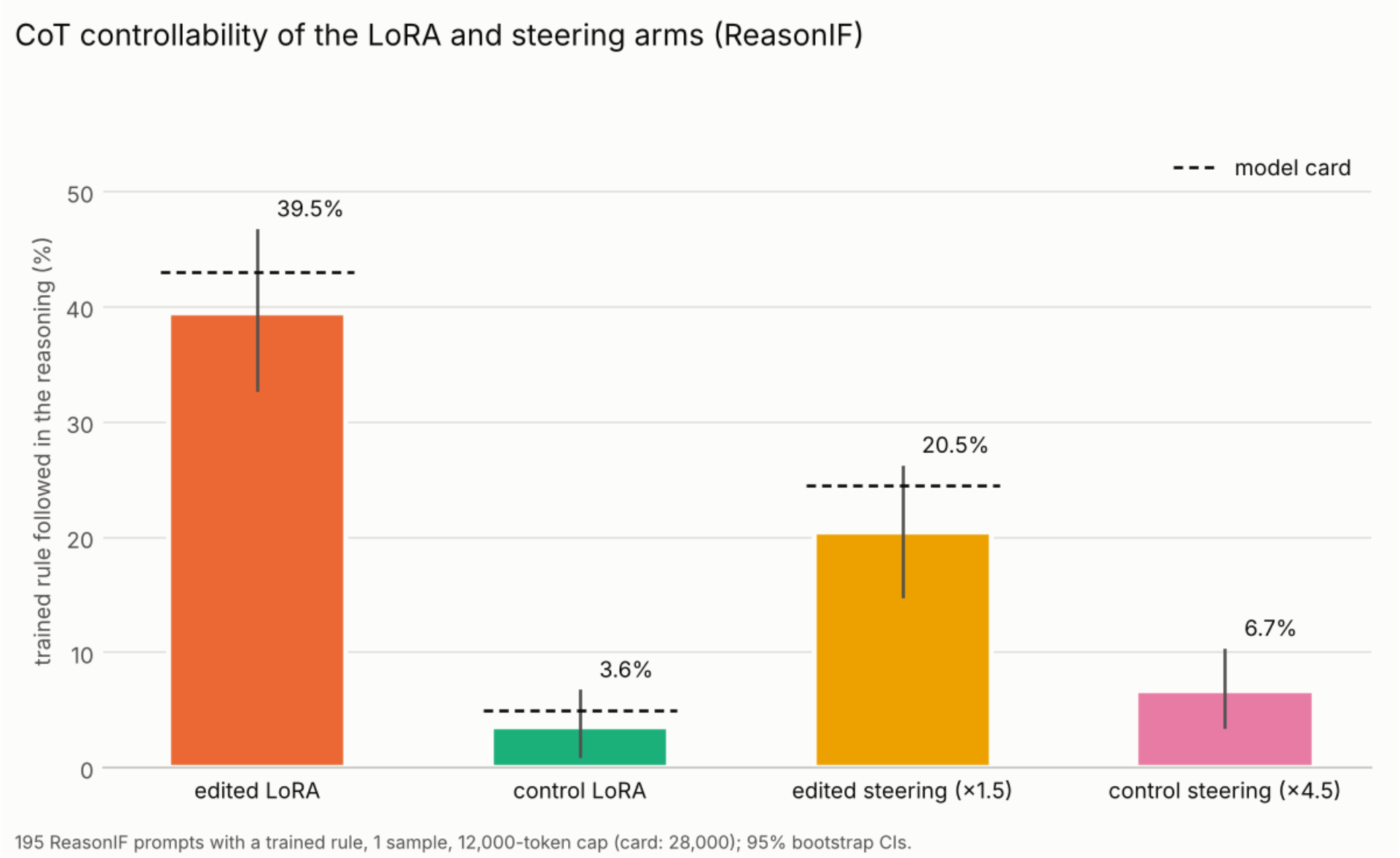
Change in verbalized awareness against the plain explicit cue (same prompt, same placement). No arm hides spontaneously under monitoring; the penalty cue and an explicit "don't mention" instruction lower it in every arm, and the CoT-controllable arms are no better at it than their controls.



Paired by prompt; 95% bootstrap CIs over prompts. Under punish, most remaining VEA quotes restate the penalty rule ("...if I mention being tested I get penalized, so I must avoid...").

Steering arms: CoT controllability

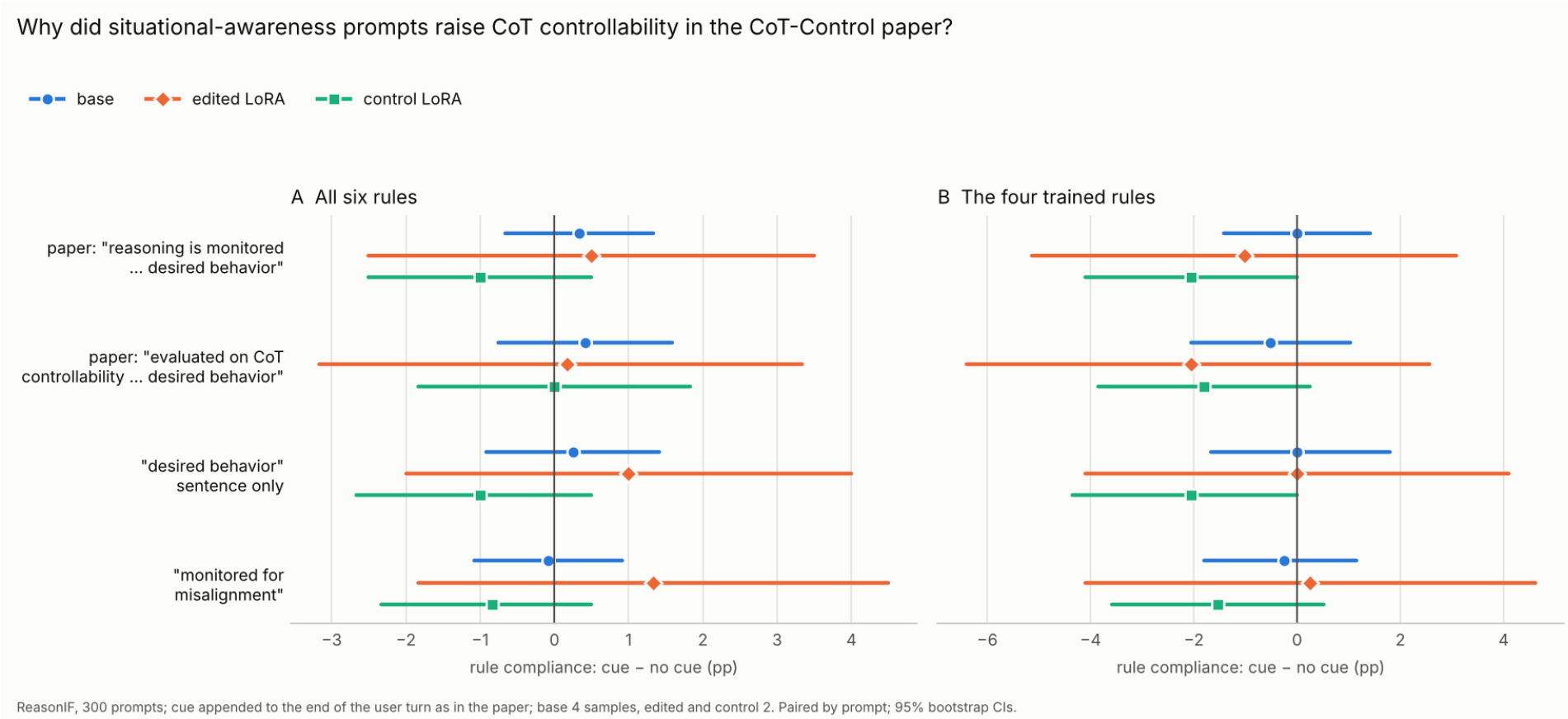
The model card's steering recipe (layer 27, edited vector ×1.5, norm-matched control ×4.5), served in vLLM with a forward hook checked against the card's Transformers hook (slope 1.00, r = 0.995). The steering gain reproduces (20.5% vs 6.7%).



195 ReasonIF prompts with a trained rule, 1 sample, 12,000-token cap (card: 28,000); 95% bootstrap CIs.

Mechanism test: the CoT-Control paper's situational-awareness prompts

Each prompt appended to the end of the user turn, as in the paper. None changes rule compliance in any arm (all within about ±2 pp), including the edited LoRA, which follows the trained rules 38% of the time with no prompt, so this is not a floor effect. The "tries harder on the task" explanation cannot be tested when the prompts have no effect to explain.



ReasonIF, 300 prompts; cue appended to the end of the user turn as in the paper; base 4 samples, edited and control 2. Paired by prompt; 95% bootstrap CIs.